```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_squared_error
```

```
df = sns.load_dataset('tips')
df.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

Next steps: Generate code with df New interactive sheet

```
y = df['tip']
X = df.drop('tip', axis=1)
```

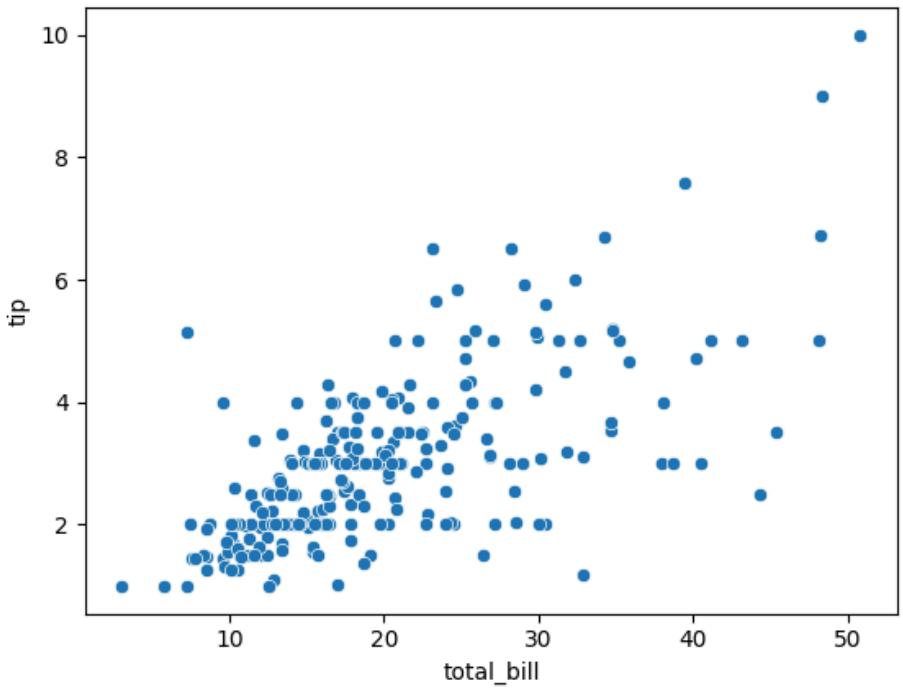
```
df.info()
df.describe()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   total_bill  244 non-null    float64
 1   tip         244 non-null    float64
 2   sex         244 non-null    category
 3   smoker      244 non-null    category
 4   day         244 non-null    category
 5   time        244 non-null    category
 6   size        244 non-null    int64
dtypes: category(4), float64(2), int64(1)
memory usage: 7.4 KB
```

	0
total_bill	0
tip	0
sex	0
smoker	0
day	0
time	0
size	0

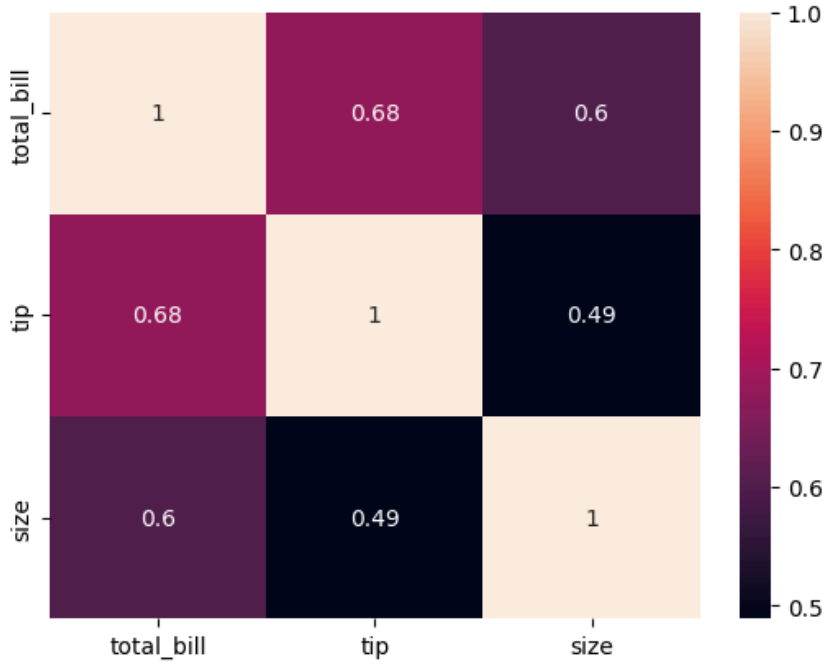
dtype: int64

```
sns.scatterplot(x='total_bill', y='tip', data=df)
plt.show()
```




```
sns.heatmap(df.corr(numeric_only=True), annot=True)
```

```
<Axes: >
```




```
cat_cols = ['sex', 'smoker', 'day', 'time']
num_cols = ['total_bill', 'size']
```

```
features = ['total_bill', 'size', 'sex', 'smoker', 'day', 'time']
X = df[features]
df
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

244 rows × 7 columns

Next steps: Generate code with df New interactive sheet

```
X_train,X_test,y_train,y_test=train_test_splitX_train,X_test,y_train,y_test=train_test_split(X,y,
```

```
preprocessor = ColumnTransformer([
('cat', OneHotEncoder(drop='first'), cat_cols),
('num', StandardScaler(), num_cols)
])
pipeline = Pipeline([
```

```
    ('preprocessor', preprocessor),
    ('model', LinearRegression())
    ])
```

```
pipeline.fit(X_train, y_train)
```

```
▸ Pipeline
  ▸ preprocessor: ColumnTransformer
    ▸ cat          ▸ num
      ▸ OneHotEncoder   ▸ StandardScaler
```